

Programme	BSc Mathematics Honours			
Course Code	MAT3VN201			
Course Title	INTRODUCTION TO MACHINE LEARNING			
Type of Course	Vocational Minor – Introduction to AI			
Semester	III			
Academic Level	200-299			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Minor 1, Minor 2 (Code)			
Course Summary	Course aims to provide basic concepts of machine learning including paradigms of supervised, unsupervised and reinforcement learning.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Machine Learning concepts and basic parameter estimation methods.	U	C	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO2	Distinguish between Supervised, Unsupervised and semi supervised learning and evaluate the performance measures	U	C	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO3	Apply the algorithms identifying problem situations	Ap	P	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Ext. Marks (70)
I		Introduction to Machine Learning	10	Min.15
	1	Introduction: Machine Learning - Machine Learning Foundations		
	2	Machine Learning Paradigms- Supervised, Unsupervised, Reinforcement		
	3	Applications of Machine Learning, Case studies		
	4	Basics of parameter estimation - maximum likelihood estimation (MLE) and maximum a posteriori Estimation (MAP).		
	5	Introduction to Bayesian formulation.		
II		Supervised Learning & SVM	14	Min.15
	6	Regression – Simple Linear regression and Multiple Linear Regression		
	7	Gradient Descent algorithm and Matrix method, Overfitting in regression.		
	8	Methods for Classification- Logistic regression, Naive Bayes, Decision tree algorithm- ID3		
	9	SVM - Introduction, Maximum Margin Classification, Mathematics behind Maximum Margin Classification		
	10	Maximum Margin linear separators, soft margin SVM classifier, non-linear SVM		
		Performance Measures & Unsupervised Learning		
	12	Regression Evaluation Metrics – Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared (Coefficient of Determination)		

III	13	Classification Evaluation Metrics - Precision, Recall, Accuracy, F-Measure, Receiver Operating Characteristic Curve (ROC), Area Under Curve (AUC)	11	Min.15
	14	Bootstrapping, Cross Validation, Ensemble methods, Bias-Variance decomposition.		
	15	Clustering - Similarity measures, Hierarchical Agglomerative Clustering, K-means partitional clustering		
	16	Expectation maximization (EM) for soft clustering		
	17	Dimensionality reduction –Principal Component Analysis, t-Distributed Stochastic Neighbour Embedding (t-SNE)		
IV		Introduction to Advanced Machine Learning	10	Min.15
	18	Introduction to Reinforcement Learning, Learning Task		
	19	Learning Models for Reinforcement – (Markov Decision process, Q Learning - Q Learning function, Q Learning Algorithm), Application of Reinforcement Learning		
	20	Introduction to Neural Network, Perceptron, Multilayer feed forward network,		
	21	Activation functions (Sigmoid, ReLU, Tanh), Back - propagation algorithm.		
	22	Case Study: Applying Reinforcement Learning in Autonomous Vehicle Navigation Case Study: Predicting Customer Churn in Telecommunications Industry using Neural Networks		
		Practical's	30	
	1	Create a dataset containing measurements of the heights of students in a class. Estimate the parameters of a normal distribution that best describes the distribution of heights using Maximum Likelihood Estimation (MLE)		

	2	The probability that it is Friday and that a student is absent is 3 %. Since there are 5 school days in a week, the probability that it is Friday is 20 %. What is the probability that a student is absent given that today is Friday? Apply Baye's rule in python to get the result		
	3	Implement Simple Linear regression using python		
	4	Implement Multiple Linear regression using python		
	5	Implement the Logistic regression algorithm		
	6	Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets		
	7	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.		
	8	Create a dataset containing information about the prices of houses in a certain city. The dataset includes various features such as the size of the house, number of bedrooms, location, and age of the house, as well as the corresponding sale prices. Your task is to build a regression model to predict the sale price of houses based on their features and evaluate the model's performance using appropriate evaluation metrics (MAE, MSE, RMSE, R-squared)		
	9	Implement the support vector machine algorithm		
	10	Create a dataset containing information about customers of a telecommunications company. The dataset includes features such as customer demographics, service usage, and contract details, as well as a binary target variable indicating whether each customer churned (1) or not (0). Your task is to build a classification model to predict customer churn based on the available features. Evaluate the trained model's performance on the testing data using the following evaluation metrics: Accuracy, Precision, Recall, F1-score and ROC Curve. Use SVM Classification		
	11	Program to implement K-Means clustering Algorithm		

	12	Create dataset containing information about customers of a retail store, including features such as age, income, and spending score. Your task is to perform clustering on the dataset to identify distinct groups of customers based on their purchasing behaviour. Use K-means Algorithm		
	13	Implement Dimensionality reduction using Principal Component Analysis (PCA) method		
	14	Implementing a simple reinforcement learning algorithm		
	15	Create a dataset containing information about patients with diabetes, including features such as age, BMI, blood pressure, and glucose levels, as well as an indication of whether each patient has diabetes or not. Your task is to build a simple neural network classifier to predict whether a patient has diabetes based on their features		
	References			
	1.	M. Gopal, “Applied Machine Learning”, McGraw Hill Education		
	2.	Tom M Mitchell, —Machine Learning, First Edition, McGraw Hill Education, 2013		
	3.	Machine Learning: A Probabilistic Perspective by Kevin P. Murphy		
	4.	Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010.		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	3	2	3	2	3	1	2
CO 2	2	3	3	2	3	2	3	1	2
CO 3	2	3	3	2	3	2	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓